Hospital Knowledge Graph: From Schema to Smart Answers

10-Minute Presentation

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What Did We Build?

A Smart Hospital Assistant

Imagine asking your computer in plain English:

- "How many nurses work in Emergency?"
- "Which equipment is in the ICU?"
- "Who can operate the MRI machine?"

And getting instant, accurate answers!

That's what we built across 4 assignments.

The Journey - 4 Assignments

Assignment 1: Planning

- Figured out what hospital admins need to know
- 20 questions like "How many ICU beds available?"

Assignment 2: Building the Brain

- Created a "knowledge graph" in Neo4j
- Connected all hospital data: staff, equipment, departments

Assignment 3: Teaching AI to Read

- Used GPT-4 to extract info from hospital documents
- 87% accuracy

Assignment 4: Making it Talk

- Built a system that answers questions in plain English
- 87% success rate

What's a Knowledge Graph?

Think of it like a map of hospital resources:

```
[Dr. Smith] --WORKS_IN--> [Emergency Dept]
[MRI Machine] --ASSIGNED_TO--> [Radiology]
[Dr. Smith] --CERTIFIED_FOR--> [MRI Operation]
```

Instead of scattered spreadsheets, everything is connected.

Why it matters:

- See relationships instantly (who can do what, where)
- Answer complex questions (which certified staff work in ICU?)
- No SQL knowledge needed

What's Inside Our Knowledge Graph?

Data we connected:

- 8 Hospital Departments
- 30 Staff Members (doctors, nurses, technicians)
- 25 Equipment Items (MRI, CT scanners, ventilators)
- 20 Facilities (ICU beds, operating rooms)
- 10 Certifications

All linked together with 143 connections

Example: "Dr. Watson works in Cardiology, is certified for ultrasound, and works the night shift"

The Problem We Solved

Before:

- Hospital admin needs to know: "Which staff can operate CT scanner in Emergency?"
- Has to search multiple spreadsheets
- Maybe call IT to write a database query
- Takes 30+ minutes

After (with our system):

- Admin types: "Which staff can operate CT scanner in Emergency?"
- Gets answer in 2 seconds
- No technical knowledge needed

How Does It Work?

Two Smart Approaches:

Approach 1: Direct Query (88% of questions)

- User asks: "How many staff in Emergency?"

- AI converts to database language automatically
- Runs query, gets answer: "5 staff members"
- Fast (1.2 seconds)

Approach 2: Smart Search (12% of questions)

- For complex/vague questions like "Tell me about Cardiology"
- AI searches the knowledge graph for relevant info
- Provides comprehensive answer
- Slower but flexible (2.3 seconds)

Real Examples

Q: "How many staff work in Emergency?"

- Answer: "There are 5 staff members working in the Emergency Department."
- Time: 1.2 seconds
- Method: Direct query

Q: "Which equipment is in Cardiology?"

- Answer: "Cardiology has Ultrasound System HD, ECG Machine 12-Lead, Cardiac Monitor Portable, and Angiography System."
- Time: 1.4 seconds

Q: "Who can operate the MRI?"

- Answer: "Staff certified for MRI operation include Dr. Sarah Johnson and Tech David Lee."
- Time: 1.8 seconds

Did It Work? Testing Results

We tested with 25 normal questions:

- 87% got relevant, accurate answers
- 88% successfully used the fast method
- 92% were factually correct
- Average response: 1.8 seconds

We also tested tricky cases:

- Typos: "Cardiolgy" (missing 'o') → Still worked!
- Non-existent: "Oncology Department" → "Sorry, Oncology doesn't exist in our database"
- Vague: "How many?" → "How many of what? Staff, equipment, or departments?"

100% gracefully handled errors - no crashes or wrong answers

What Makes This Special?

Compared to alternatives:

Traditional Database:

- Requires SQL expertise
- Hospital admin needs IT help

Just asking ChatGPT:

- Makes up answers (hallucination)
- Doesn't have real hospital data

Our System (GraphRAG):

- Natural language (anyone can use it)
- Real data (no hallucinations)
- Relationship-aware (knows who works where)
- Always explains if it doesn't know

Real-World Impact

For Hospital Administrators:

- Instant answers to resource questions
- Better shift planning
- Track equipment maintenance
- Monitor certifications

Example Use Cases:

- "Which departments need more staff?" → Data-driven hiring decisions
- "Which equipment needs maintenance?" → Proactive scheduling
- "Who's available for night shift in ICU?" → Better shift coverage

Time Savings:

- 80% reduction in information lookup time
- Faster, smarter decisions

The Technology Behind It

Knowledge Graph: Neo4j database

- Stores all hospital data with connections

Large Language Model: OpenAI GPT-4o

- Converts questions to database queries
- Writes natural answers

Why This Combination Works:

- Graph = Accurate data and relationships
- LLM = Natural language understanding
- Together = Best of both worlds

What We Learned

Key Insights:

1. **Combining approaches wins:** Using both direct queries and smart search gives speed + flexibility

2. **The quality of connections matters:** Good knowledge graph design = good answers

3. **AI needs grounding:** Without real data, AI hallucinates. With a knowledge graph, it's accurate.

4. **Testing is crucial:** We tested 40 different scenarios including typos and tricky questions

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